

# Feature Ticket List

PrepFlow — Hybrid AI Exam Practice & Progress Tracker

## Purpose

This is the build checklist. Tickets are ordered by dependency and are written so each can be used as a focused prompt for an AI coding tool. Priorities: Must-have = required for MVP launch; Should-have = post-core improvement; Nice-to-have = later enhancement.

## Epic A — Foundation

### PF-001: Initialize application — Must-have

**Description:** Create Next.js + TypeScript + Tailwind project, strict type checking, clean folders, linting, .env.example.

**Acceptance criteria:** App runs locally; lint/type checks pass; no real secrets committed.

**Dependencies:** None

### PF-002: Configure Supabase — Must-have

**Description:** Connect Auth, PostgreSQL, and Storage with distinct browser/server clients.

**Acceptance criteria:** Secrets stay server-only; missing env vars fail clearly.

**Dependencies:** PF-001

### PF-003: Create schema/migrations — Must-have

**Description:** Create profiles, exams, books, sections, questions, sessions, attempts, goals, revisions, jobs, and audit logs.

**Acceptance criteria:** Foreign keys, enums, indexes, constraints and clean migration work.

**Dependencies:** PF-002

### PF-004: Enable RLS policies — Must-have

**Description:** Apply ownership rules across user data.

**Acceptance criteria:** Cross-user reads/writes are denied in tests.

**Dependencies:** PF-003

### PF-005: Private storage bucket — Must-have

**Description:** Create user-scoped private books bucket and policies.

**Acceptance criteria:** No public URLs; owner-only object access; signed downloads.

**Dependencies:** PF-002

### PF-006: Design-system foundation — Must-have

**Description:** Implement colors, typography, buttons, inputs, cards, badges, modals, error/empty states.

**Acceptance criteria:** Responsive, accessible components with focus states.

**Dependencies:** PF-001

### PF-007: Signup and verification — Must-have

**Description:** Email/password signup and verification.

<b>Acceptance criteria:</b> Profile created safely; verification required.

<b>Dependencies:</b> PF-002, PF-003, PF-004, PF-006

### **PF-008: Login/reset/logout — Must-have**

<b>Description:</b> Secure auth pages and protected routing.

<b>Acceptance criteria:</b> Safe errors, redirects, rate-limited auth paths.

<b>Dependencies:</b> PF-007

### **PF-009: Onboarding — Must-have**

<b>Description:</b> Collect exam, date, timezone, daily target.

<b>Acceptance criteria:</b> Valid positive targets; setup completion saved.

<b>Dependencies:</b> PF-007

### **PF-010: Secure book upload — Must-have**

<b>Description:</b> Upload PDF/DOCX/TXT to private storage and create book/job records.

<b>Acceptance criteria:</b> File validation, progress, consent, safe path.

<b>Dependencies:</b> PF-005, PF-009

## **Epic B — Document processing**

### **PF-011: Processing job framework — Must-have**

<b>Description:</b> Queue extraction, classification, indexing, cleanup, retry jobs.

<b>Acceptance criteria:</b> Durable statuses, safe errors, retries, polling UI.

<b>Dependencies:</b> PF-010

### **PF-012: Text extraction/OCR detection — Must-have**

<b>Description:</b> Extract text and detect scanned/unreadable content.

<b>Acceptance criteria:</b> Page boundaries recorded; protected/unsupported files handled.

<b>Dependencies:</b> PF-011

### **PF-013: Document classifier — Must-have**

<b>Description:</b> Classify as question bank, theory, mixed, or unknown.

<b>Acceptance criteria:</b> Confidence stored; user can override uncertain result.

<b>Dependencies:</b> PF-012

### **PF-014: Question-bank extraction — Must-have**

<b>Description:</b> Extract ordered MCQs, A-D, answer keys, explanations, source pages.

<b>Acceptance criteria:</b> Invalid/missing fields flagged; order preserved; schema validation.

<b>Dependencies:</b> PF-012, PF-013

### **PF-015: Question review screen — Should-have**

<b>Description:</b> Let user review/edit flagged extracted questions.

<b>Acceptance criteria:</b> Approved items enabled; edits audited; rejected items hidden.

<b>Dependencies:</b> PF-014

### **PF-016: Theory retrieval indexing — Must-have**

<b>Description:</b> Create per-book retrieval/vector-store index.

<b>Acceptance criteria:</b> Ready state only after indexing; ownership isolation; cleanup on delete.

<b>Dependencies:</b> PF-012, PF-013

### **PF-017: Mixed section mapping — Should-have**

<b>Description:</b> Map chapters as theory/question-bank/mixed.

<b>Acceptance criteria:</b> Sections link to pages/questions; user can correct labels.

<b>Dependencies:</b> PF-014, PF-016

## **Epic C — Practice engine**

### **PF-018: Practice-session lifecycle — Must-have**

<b>Description:</b> Create/resume/end primary, revision, and topic sessions.

<b>Acceptance criteria:</b> Owner-only sessions; counters/timestamps persist.

<b>Dependencies:</b> PF-003, PF-004, PF-007

### **PF-019: Next extracted question — Must-have**

<b>Description:</b> Return next valid unanswered MCQ in source order.

<b>Acceptance criteria:</b> No key before submit; correct completion state; owner/ready check.

<b>Dependencies:</b> PF-014, PF-018

### **PF-020: Theory question generation — Must-have**

<b>Description:</b> Retrieve source, create one valid MCQ, validate and persist before display.

<b>Acceptance criteria:</b> Four unique options; one key; source reference; duplicate prevention.

<b>Dependencies:</b> PF-016, PF-018

### **PF-021: Practice question UI — Must-have**

<b>Description:</b> Build mobile-first A-D selection, timer, unsure, submit.

<b>Acceptance criteria:</b> Accessible keyboard/touch selection and responsive layout.

<b>Dependencies:</b> PF-006, PF-018, PF-019

### **PF-022: Secure answer scoring — Must-have**

<b>Description:</b> Server scores and saves attempts/revisions.

<b>Acceptance criteria:</b> Idempotent; server-owned key; attempt/goal/revision updates consistent.

<b>Dependencies:</b> PF-021, PF-003, PF-004

### **PF-023: Post-answer feedback — Must-have**

<b>Description:</b> Show result, key, explanation, source, revision status, next action.

<b>Acceptance criteria:</b> Supportive, clear and accessible states.

<b>Dependencies:</b> PF-022

### **PF-024: Skip/unsure handling — Must-have**

<b>Description:</b> Record nonstandard outcomes without corrupting progress.

<b>Acceptance criteria:</b> Explicit metrics and correct revision/sequence behavior.

<b>Dependencies:</b> PF-022

## Epic D — Progress and revision

### PF-025: Daily goals/timezone tracking — Must-have

<b>Description:</b> Calculate today/yesterday based on user timezone.

<b>Acceptance criteria:</b> Counts derive from attempts; midnight tests pass.

<b>Dependencies:</b> PF-022, PF-009

### PF-026: Dashboard overview — Must-have

<b>Description:</b> Show daily target, yesterday, next action, revision, progress, accuracy.

<b>Acceptance criteria:</b> Live post-answer update and useful empty states.

<b>Dependencies:</b> PF-025, PF-019, PF-020, PF-023

### PF-027: Revision scheduler — Must-have

<b>Description:</b> Schedule wrong/unsure items at 1,3,7,14,30-day intervals.

<b>Acceptance criteria:</b> No duplicate active queue entries; mastery updates.

<b>Dependencies:</b> PF-022

### PF-028: Revision practice — Must-have

<b>Description:</b> Practice due revisions with same scoring flow.

<b>Acceptance criteria:</b> Revision mode is stored and does not alter primary sequence.

<b>Dependencies:</b> PF-027, PF-021, PF-022

### PF-029: Streaks — Should-have

<b>Description:</b> Calculate active and goal-met streaks.

<b>Acceptance criteria:</b> Timezone-aware, correct gap behavior.

<b>Dependencies:</b> PF-025

## Epic E — Analytics, operations, launch

### PF-030: Analytics calculations — Must-have

<b>Description:</b> Compute volume, accuracy, completion, subject/chapter and revision performance.

<b>Acceptance criteria:</b> Derived from attempts; indexed queries; clear metric definitions.

<b>Dependencies:</b> PF-022, PF-025, PF-027

### PF-031: Analytics UI — Must-have

<b>Description:</b> Build readable volume/accuracy/topic/revision charts.

<b>Acceptance criteria:</b> Text summaries, filters, mobile-safe layout.

<b>Dependencies:</b> PF-030, PF-006

### PF-032: Book management — Must-have

<b>Description:</b> List, set active, inspect, retry, download, delete books.

<b>Acceptance criteria:</b> Owner-only; safe deletion and cleanup.

<b>Dependencies:</b> PF-010, PF-011, PF-016

### PF-033: Data export — Should-have

<b>Description:</b> Export own attempts/progress/revisions as CSV.

<b>Acceptance criteria:</b> Secure, audited, own-data only.

<b>Dependencies:</b> PF-030, PF-004

### **PF-034: Rate limiting — Must-have**

<b>Description:</b> Protect auth, uploads, AI generation, hints, and submissions.

<b>Acceptance criteria:</b> User-friendly limit errors and safe logs.

<b>Dependencies:</b> PF-008, PF-010, PF-020

### **PF-035: Audit logging — Must-have**

<b>Description:</b> Log uploads, deletes, exports, admin/support actions.

<b>Acceptance criteria:</b> No secret/raw content in logs; append-only behavior.

<b>Dependencies:</b> PF-003, PF-004

### **PF-036: Error monitoring — Must-have**

<b>Description:</b> Add monitoring and user-safe error boundaries.

<b>Acceptance criteria:</b> Sanitized context, retry UI, job visibility.

<b>Dependencies:</b> PF-001, PF-011

### **PF-037: Admin operations dashboard — Should-have**

<b>Description:</b> Show job health and aggregate metrics without private-content access.

<b>Acceptance criteria:</b> Admin-only and audited.

<b>Dependencies:</b> PF-004, PF-011, PF-035

### **PF-038: Deletion lifecycle — Must-have**

<b>Description:</b> Delete book/account data across storage, DB, and retrieval resources.

<b>Acceptance criteria:</b> Immediate user hiding, retryable cleanup, confirmation.

<b>Dependencies:</b> PF-005, PF-011, PF-016, PF-035

### **PF-039: Core unit/integration tests — Must-have**

<b>Description:</b> Test selection, scoring, duplicates, revision and RLS.

<b>Acceptance criteria:</b> Tests run in CI and cover cross-user denial.

<b>Dependencies:</b> PF-019, PF-020, PF-022, PF-027

### **PF-040: End-to-end tests — Must-have**

<b>Description:</b> Test signup→upload→practice→dashboard→revision.

<b>Acceptance criteria:</b> Desktop/mobile workflows pass.

<b>Dependencies:</b> PF-039, PF-026, PF-028

### **PF-041: Accessibility/responsive audit — Must-have**

<b>Description:</b> Keyboard, focus, contrast, charts, responsive tests.

<b>Acceptance criteria:</b> WCAG AA-oriented checks and small-device support.

<b>Dependencies:</b> PF-006, PF-021, PF-026, PF-031

### **PF-042: Deploy staging/production — Must-have**

<b>Description:</b> Configure Vercel, env separation, migrations, monitoring, health check.

<b>Acceptance criteria:</b> Documented rollback and backups.

<b>Dependencies:</b> PF-036, PF-039, PF-040

## Recommended release order

1) PF-001–010 foundation/auth/upload; 2) PF-011–016 processing and content preparation; 3) PF-018–024 practice/scoring; 4) PF-025–028 goals/dashboard/revision; 5) PF-030–032 analytics/book management; 6) PF-034–042 hardening, tests, deployment. Add review screens, detailed section mapping, exports, streaks, and admin operations after the core daily practice loop is stable.